Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python
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Education

Mathematics Modules (NetMath) — Online (Urbana-Champaign, USA)
University of Illinois Urbana-Champaign — Apr 2026 – Dec 2026

- Credit-bearing, exam-assessed proof-based mathematics, studied alongside the MSc.
- **MATH 314** Introduction to Higher Mathematics *(in progress, 100%)* — logic, set theory, ε-δ arguments, proof techniques.
- **MATH 447** Real Analysis *(upcoming)* — sequences & convergence, metric spaces, compactness, continuity, differentiation, Riemann integration.

MSc Scientific Computing & Data Analysis (AI for Engineering) — Durham, United Kingdom
Durham University — Sep 2025 – Exp. Sep 2026

- Tracking Distinction — average ∼86% (95% performance/GPU, 94% machine learning & statistics).
- **Dissertation:** Deep BSDEs for mean-field market making (Cont–Xiong), with Lean-verified error certification.
- **Focus:** Bayesian inference, convex optimisation, deep learning, HPC/GPU. Python, C, R.

MEng Electronic Engineering — Durham, United Kingdom
Durham University — Sep 2020 – Jun 2024

- Upper Second-Class Honours (2:1).
- **Dissertation:** *Silicon Carbide JFET CPU* — simulation-driven CPU design with instruction-level verification.

Experience

Industrial Tutor — Durham, United Kingdom
Durham University, Department of Engineering — Sep 2025 – Present

- Led weekly design tutorials, mentoring teams on feasibility-driven decision-making under cost, risk, and performance constraints.
- Assessed and provided structured feedback on technical feasibility, assumptions, and justification quality.

Graduate Communications Engineer — Hebburn, United Kingdom
Siemens PLC — Sep 2024 – Present

- Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools.
- Developed a Python-based RAG pipeline to structure and query internal technical documentation and codebases using a retrieval-augmented workflow (local LLMs, vector search) reducing engineer time to answer by >90%.

Projects & Research

Deep BSDEs for Mean-Field Market Making — Durham, United Kingdom
Durham University — 2026 – MSc Dissertation

- Built deep backward-SDE-with-jumps solvers for a competitive mean-field market-making game (Cont–Xiong), validated to 0.135% spread error vs an exact benchmark and scaled to ∼5,000 agents.
- Showed via a 20-seed multi-agent-RL study (MADDPG) that learning algorithms reach **supra-cartel** spreads (t=3.22, p=0.0022) — tacit collusion is a property of the learning dynamics, not the market equilibrium.
- Developed an a-posteriori error-certification theory for deep-BSDE solvers and machine-checked its core in Lean 4 / Mathlib (∼36k LOC, axiom-clean).

Silicon Carbide JFET CPU — Durham, United Kingdom
Master's Dissertation — Oct 2023 – Apr 2024

- Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments.

- Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction.
- Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.

Certifications

- MITx 15.455x – Mathematical Methods for Quantitative Finance (In Progress)
- Columbia University – Introduction to Financial Engineering and Risk Management

Leadership, Activities & Interests

Research Interests: Time-series modelling, signal validation, regime dependence in financial systems.
Leadership: Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).
Activities: Durham University Fencing Team; Boxing (Student Fight Night).
Achievements: South Africa U17 Fencing Champion & Weldon le Huray Scholar; President's Award (Gold).

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Python; C++; High-Performance Computing; Linear Algebra